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Measuring thresholds for individual change can improve social change interventions

Behavioural and complexity sciences have largely proceeded separately in studying social change. By reinterpreting social contagion dynamics through discrete-choice modelling, we demonstrated how thresholds for individual change can be estimated from choice data, and theoretically examined how they can improve interventions for social change.

This is a summary of:
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The problem

Desired behavioural changes for mitigating global challenges are often well known, yet they face persistent social and political resistance. For example, this paradox is evident in the climate crisis: despite evidence on how shifts in mobility, building design and energy policies can reduce global emissions, support for climate policies and the adoption of sustainable behaviours often remain limited[1]. In such high-resistance contexts, complex contagion theory — the leading complexity science framework for modelling how behavioural change spreads in social networks — posits that individuals require social reinforcement from their peers before adopting a new behaviour. This mechanism is captured by an individual-level trait known as the adoption threshold, which quantifies the minimal level of social reinforcement needed before a new behaviour is adopted[2]. Yet there has been a lack of empirical measurements of individual adoption thresholds. Early works suggested that these could directly inform interventions for social change (for example, by revealing low-threshold individuals who might be engaged to accelerate the adoption of behavioural change)[3]. However, because of the lack of validated adoption-threshold measurements, it remains unclear in what conditions and to what extent individual adoption thresholds can improve interventions for social change.

The solution

Reinterpreting individual adoption thresholds in terms of discrete-choice models came to the rescue. Our work started by analytically deriving individual adoption thresholds from a utility-based discrete-choice model from the consumer behaviour literature[4]. We then designed two choice-based conjoint experiments (surveys in which researchers observe how respondents choose among options with varying attributes) that covered two high-resistance contexts. In the two experiments, participants had to repeatedly choose which energy policies to support and which new instant messaging apps to adopt, respectively. Crucially, through this conjoint design, we could expose participants to varying degrees of social support for each energy policy or app alternative. From the collected data, we estimated individual-level adoption thresholds for each energy policy and each app. We subsequently performed extensive data-driven simulations to determine whether incorporating the estimated adoption thresholds into social change interventions improved behavioural change outcomes as compared to traditional interventions. We focused on seeding interventions: social change interventions that accelerate large-scale behavioural change by initially incentivizing a small number of 'seed individuals' to adopt.

We found that social change interventions informed by both social network structure and behavioural traits markedly outperform state-of-the-art interventions that consider only social network structure or only behavioural traits (Fig. 1a). Initially targeting social network nodes connected with many low-threshold nodes proved to be particularly effective when the cost of successfully targeting a node only depended on an individual's preference for the new behavioural choice. By contrast, in scenarios in which the cost of targeting grows with the popularity of the social network node (as in online influencer marketing), the best interventions relied on complex centrality[5], which accounts for adoption thresholds and network paths beyond a node's immediate social connections (Fig. 1b). Hence, integrating estimated thresholds into the social change intervention proved to be essential for improving outcomes.

The implications

Broadly, our findings suggest that the integration of models from the behavioural and complexity sciences can substantially improve the design of social change interventions. As change results from both individual and social drivers, behavioural models enable data-driven individual-level modelling whereas complexity science models clarify how individual changes propagate through networks. We envision that future efforts could extend the proposed integration beyond the discrete-choice experiments and social contagion models studied in our work.

A limitation of the developed experimental design is that it relies on simple social cues without capturing the rich spectrum of social signals and social roles encountered in real-world contexts. Further, in real-world contexts, individual preferences and social cues might co-evolve, which is not captured by our modelling framework.

Natural extensions of this work involve capturing more realistic social cues, roles and feedback mechanisms in the experimental design and behavioural modelling. Critically, field experiments are needed to validate the resulting theoretical findings, probe their boundaries and contribute to positive change in the real world.

Manuel S. Mariani & Radu Tănase
University of Zurich, Zurich, Switzerland.

EXPERT OPINION

"The manuscript addresses an important and timely topic in behavioural and computational social science, focusing on the feasibility of integrating individual preferences into collective behaviour models. The work is particularly valuable for its empirical approach, as it tests whether it is possible, and how, to derive threshold distributions from data." **Vítor V. Vasconcelos, Computational Science Lab, University of Amsterdam, Amsterdam, Netherlands.**

FIGURE

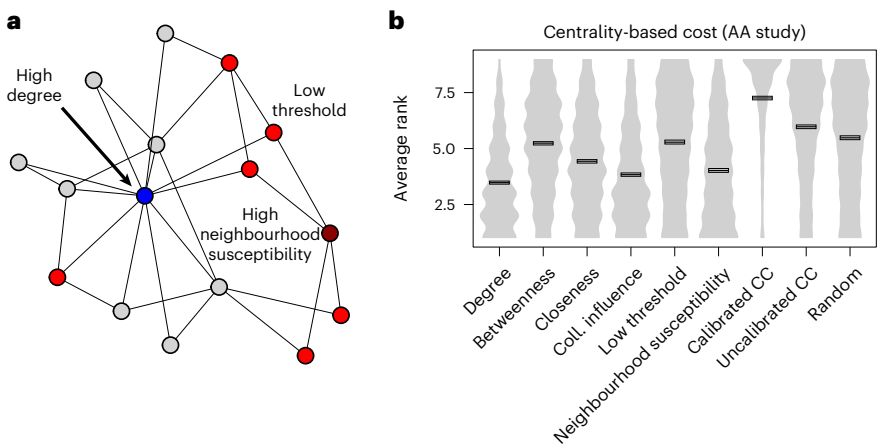

Fig. 1 | Relative performance of social change interventions. We compare 'integrative' seeding interventions informed by both individual-level adoption thresholds and social network structure with interventions informed by only one of these sources. **a**, Network-only interventions rely on social network structure to identify seed nodes who are initially incentivized to adopt (for example, the highest-degree node in blue). Behaviour-only interventions select seeds on the basis of their inferred preferences (for example, a low-threshold node in red). Integrative interventions select seeds on the basis of both of these data sources (for example, the dark red node, which has the largest number of connections to low-threshold nodes). **b**, Relative performance of seeding interventions that differ in the seed selection criterion under a centrality-based cost structure for the app adoption (AA) study, showing the performance advantage of an integrative seeding intervention: complex centrality (CC) calibrated with the estimated thresholds. Coll., collective. © 2026, Tănase, R. et al.

BEHIND THE PAPER

An important 'a-ha' moment was the realization that conjoint experiments frequently used in marketing science provide a natural setup for estimating adoption thresholds. Once the estimation problem was solved, we could proceed with using the estimated thresholds to inform interventions. We arrived at the idea of using the estimated thresholds as inputs to complex contagion models only after extended conversations with colleagues and frequent exchanges with researchers across diverse fields, including marketing science, sociology, development economics and network science. Interdisciplinary exchanges were key to our paper. **M.S.M. & R.T.**

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FROM THE EDITOR

"The adoption of new products, ideas and behaviours can spread like a 'contagion' through social networks, but individuals differ in their tendencies to adopt particular innovations. Tănase et al. tackled the understudied question of how to estimate individuals' adoption tendencies." **Jamie Horder, Senior Editor, *Nature Human Behaviour*.**